
Water Treatment & Cooling Water SOP — Bioethanol Plant (excerpt)

Supplied grounding document for capstone groups (e.g. a water-treatment agent) that don't have their own. Covers the cooling water loop to condenser C-201 and the site water treatment plant W-301.

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1. Purpose & scope

Defines normal operation and monitoring of the cooling water loop serving condenser C-201, and the boiler feed / process water treatment plant W-301. Applies to utilities operators and shift engineers.

2. Cooling water loop (to condenser C-201)

Parameter	Normal	Alarm	Notes
Supply temperature	22–28 °C	> 32 °C	Above this, condenser duty and column top pressure (PT-201) can be affected
Return temperature	≤ 38 °C	> 42 °C	High rise indicates fouling or low flow
Flow to C-201	Per design duty	< 90% of design	Check strainer, pump performance
Conductivity (cycles of concentration)	Per cooling tower dosing plan	Outside dosing plan band	Indicates blowdown/make-up imbalance

If cooling water supply temperature rises toward the alarm band, condenser duty falls and column top pressure (PT-201, SOP-D201-014 §6) can trend upward — check the cooling tower and C-201 cooling water flow before adjusting the column.

3. Water treatment plant W-301 (boiler feed / process water)

Parameter	Target	Alarm
Feed water hardness	< 1 ppm (as CaCO ₃)	> 2 ppm
Feed water pH	8.5–9.5	Outside band
Dissolved oxygen (deaerator outlet)	< 10 ppb	> 20 ppb
Conductivity	Per boiler manufacturer limit	Above limit

Poor boiler feed water quality is a leading cause of boiler tube scaling and corrosion — see the Steam & Utilities SOP for boiler-side limits and escalation.

4. Sampling & routine checks

- Cooling tower basin and boiler feed water are sampled each shift; results are logged against the tables above.
- A single out-of-band reading should be re-checked before action; a **sustained** trend across two consecutive shifts should be escalated.
- Biocide and scale-inhibitor dosing changes follow the site water treatment chemical program — do not adjust dosing rates without utilities-engineer approval (dosing chemicals carry their own SDS; treat with normal PPE for

chemical handling).

5. Escalation & human sign-off

- Cooling water or W-301 parameters trending into the alarm bands should be flagged to the shift utilities engineer — an assistant may recommend likely causes and cite this SOP, but must not change dosing rates, blowdown rates, or cooling tower fan/pump duty directly.
- Any water treatment issue that could affect boiler safety (§3) or discharge quality (see Effluent Compliance Standard) requires engineering sign-off before corrective action.